Sub Code:NME701

Paper Id: 140701 Roll No:

B. TECH
(SEM-VII) THEORY EXAMINATION, 2019-20
CAD

Time: 3 Hours *Total Marks: 100*

Note: Attempt all Sections. If require any missing data; then choose suitably.

SECTION A

1. Attempt *all* questions in brief. 2 x 10 = 20

a. Define emissive and non-emissive display.
b. What are the essential requirements of CAD?
c. What are the basic rules that are considered before designing graphics software?
d. Write down 2D transformation matrix for Rotation and Reflection about origin.
e. What are essential requirements for synthetic curves?
f. Write any four properties of Bezier curve.
g. Differentiate between plane surface and ruled surface with neat sketch.
h. What are the limitations of wireframe modeling?
i. State the principal of minimum potential energy.
j. How element type and size play an important role for mesh generation?

www.aktuonline.com

SECTION B

2. Attempt any *three* of the following: 10 x 3 = 30

a. What do you understand by computer Integrated Manufacturing (CIM)? Discuss its role or function in product development cycle with suitable block diagram.
b. Explain the DDA line generation techniques for line with different slopes. What are the limitations of DDA algorithms?
c. Prove that the Bezier polynomials obtained from Hermite form and that obtained from Bernstein polynomials are same.
d. Explain constructive solid geometry. What is the role of primitives and Boolean operations in CSG? Explain with suitable examples.
e. Using variational approach drive the element stiffness matrix for one dimensional bar element.

SECTION C

3. Attempt any *one* part of the following: 10 x 1 = 10

(a) Write short notes on followings:
(i) Light pen
(ii) Digitizer
(b) Discuss the differences between beam penetration method and shadow mask method for obtaining colour pictures on CRT display.

4. Attempt any *one* part of the following: 10 x 1 = 10

(a) Explain Bresenham's line algorithm and digitize the line with end points A (18, 10) and B (23, 22), also plot a pixel position using Bresenham's line algorithm.
(b) Find the transformed coordinates of a triangle having vertices A (4,1), B(7,1) and C(7,3) subjected to reflection through the line $2y = x+2$.

http://www.aktuonline.com

5. Attempt any *one* part of the following: 10 x 1 = 10

(a) What do you understand by the interpolation spline and approximation splines? Explain each with suitable examples.

(b) Generate parametric equation of a three-dimensional Bezier curve defined by the four-control points P_o (5, 4, 2), P_1 (6, 2, 3), P_2 (5, - 2, 4) and P_3 (6, - 4, 2).

6. Attempt any *one* part of the following: 10 x 1 = 10

(a) Describe the following techniques used in solid modeling packages:
(i) Sweep representation
(ii) Boundary representation

(b) What is colour model? Differentiate between RGB and CYMK colour models.

7. Attempt any *one* part of the following: 10 x 1 = 10

(a) A force of 50 kN is applied at a connecting end of two-bar truss as shown in fig.1. Determine the reactions, stresses and strains on each member of truss. Take E = 180 GPa, A = 200 mm^2.

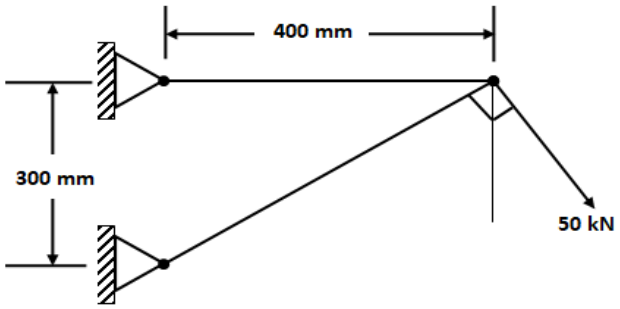

Figure.1

(b) What do you understand by Local coordinates, Global coordinates and Natural coordinates in FEM? Explain their utility with example.

www.aktuonline.com